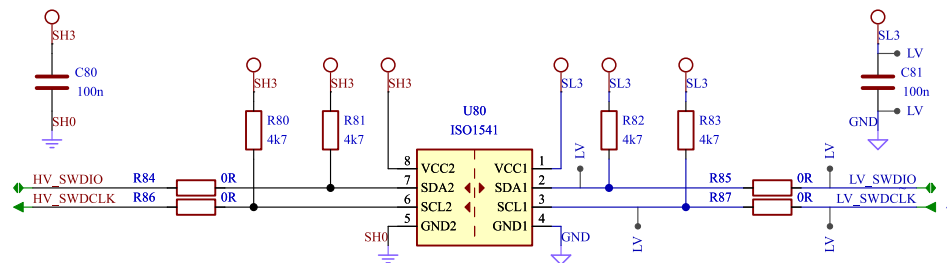
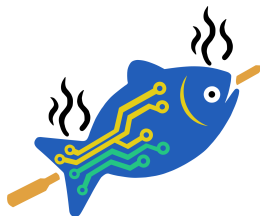


Pin connection diagram for the HV module. The diagram shows a top header with pins SH3, SL3, and SH0. SH3 is connected to HV_SWDIO. SL3 is connected to HV_SWDCLK. SH0 is connected to HV_EXT_EN. The bottom header has pins LV_SWDIO, LV_SWDCLK, LV_NRST, LV_TX, LV_RX, LV_EXT_EN, and GND. LV_SWDIO is connected to HV_SWDIO. LV_SWDCLK is connected to HV_SWDCLK. LV_NRST is connected to HV_NRST. LV_TX is connected to HV_TX. LV_RX is connected to HV_RX. LV_EXT_EN is connected to HV_EXT_EN. GND is connected to GND.



The schematic diagram illustrates the internal circuitry of the HV module, centered around the ISO6741 chip (U81). The chip is configured with its V_{cc} pins (16 and 1) connected to the SH3 input through a 10k resistor (R88). The GND pins (9 and 2) are connected to the SH0 input through a 10k resistor (R98). The output pins (14, 13, 12, 11, 10) are connected to the HV module pins (R90, R92, R94, R96) through 0R resistors. The input pins (3, 4, 5, 6, 7) are connected to the HV module pins (R91, R93, R95, R97) through 0R resistors. The chip is also connected to the LV module pins (LV TX, LV NRST, LV EXT_EN, LV RX) through 0R resistors. The chip is powered by a 100nF capacitor (C82) connected to SH3 and SH0. The chip is also connected to the LV module pins (LV TX, LV NRST, LV EXT_EN, LV RX) through 0R resistors. The chip is also connected to the LV module pins (LV TX, LV NRST, LV EXT_EN, LV RX) through 0R resistors.

Optical isolation



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Electronics

Comments:	Company:		Variant: CHECKED	Git Hash: 84f4bdf
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Optical isolation	File Name: Optical_isolation.kicad_sch	Designer: FR	Date: 2025-07-04	Revision:
Sheet Path: /Block Diagram/Optical isolation/		Reviewer:	Size: A4	Sheet: 7 of 8